# Wearable Technology: Fatigue and Drowsiness Risk Detection

Project report | Muhammad Anas Murtaza | Python, scikit-learn, tkinter | Use case: workplace safety and driver-fatigue research

## 1. Purpose

Identify **possible fatigue risk** (Low / Moderate / High) from **heart rate (with HRV), movement and sleep-related data**, plus the work schedule (hours awake, hours on shift, time of day). A supervisor or driver gets the risk level, the factors behind it and a recommended action. It is a research prototype for decision support, **not** a medical or safety device.

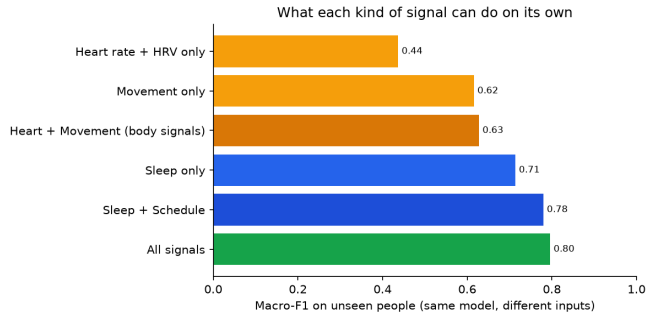
## 2. Method

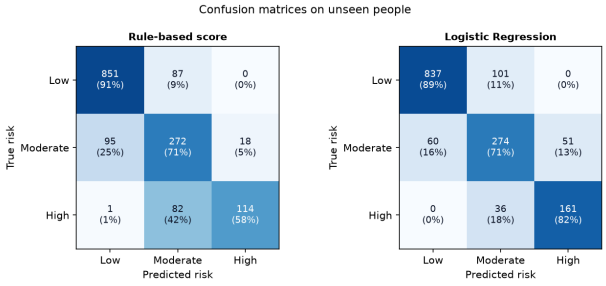
**Data:** no real wearable recordings were available, so a **simulated** dataset was generated: 150 people x 40 monitoring windows = 6,000 rows (day, evening and night shifts; personal baselines for resting HR and HRV). A hidden fatigue score built from sleep, sleep debt, hours awake, time on shift and the body-clock low points (about 03-05 h and 14-16 h) makes the body signals react (HRV falls, heart rate rises, movement drops) with heavy noise. Labels: Low 59 %, Moderate 28 %, High 13 %. **Features:** 13, including baseline-relative ones (HR above own resting rate, HRV vs own normal) and a circadian-risk curve. **Models:** an explainable rule-based score (baseline), Logistic Regression, Random Forest and Gradient Boosting. **Evaluation:** split **by person** (112 training / 38 test people, nobody in both), model chosen by 5-fold grouped cross-validation on the training people only.

## 3. Results on unseen people (1,520 test windows)

| Model | CV macro-F1 | Accuracy | Macro-F1 | Recall of High | Precision of High |
|---|---|---|---|---|---|
| Rule-based score (baseline) | - | 0.814 | 0.752 | 0.579 | 0.864 |
| **Logistic Regression (chosen)** | **0.799** | **0.837** | **0.796** | **0.817** | **0.759** |
| Random Forest | 0.793 | 0.846 | 0.794 | 0.721 | 0.807 |
| Gradient Boosting | 0.787 | 0.843 | 0.791 | 0.726 | 0.822 |

The ML model finds **82 %** of High-risk windows against 58 % for the rule baseline. Missing a fatigued person is the costly error in safety work, so recall of High is the key metric. The models are close to each other (0.79-0.80 macro-F1), so the simple, explainable Logistic Regression was kept.




## 4. Findings

- **Sleep and schedule carry most of the information** (sleep only: macro-F1 0.71; sleep + schedule: 0.78; all signals: 0.80).
- **Heart rate and HRV alone are weak** (0.44) and **movement alone is moderate** (0.62); together 0.63. Combining sources is the sensible design.
- High risk is concentrated in **night-shift** work (35 % of night windows vs 4 % of day windows) and at the body-clock low points.

## 5. How to run

`pip install -r requirements.txt`, then `python src/generate_data.py`, `python src/train_model.py` (optional - files are included) and `python src/fatigue_app.py` for the desktop app. `fatigue_risk_analysis.ipynb` holds the full analysis with outputs. Packages: numpy, pandas, scikit-learn, matplotlib, joblib (tkinter ships with Python).

## 6. Limitations and ethics

**The data is simulated, so the accuracy shown says the pipeline works, not how well it would work on real people**; labels come from a model of fatigue, not measured drowsiness, so the model partly learns the simulator's rules. Next step: validate on real wearable data against a reference such as a reaction-time test (PVT) and calibrate per person. Physiological data of workers is sensitive: use needs consent and data minimisation, and the output must support safety measures (rest rules, supervisors), never be used for discipline or surveillance, and never be the only barrier.